

Investigating the Effects of Pilates on Body Composition, Posture, and Psychosocial Parameters in Women With and Without Obesity: A Randomized Controlled Study

Sema Arslan Kabasakal¹ and Binnur Delice²

¹Department of Sports Health Science, Faculty of Sports Science, Yalova University, Yalova, Turkey;

²Department of Coaching Education, Faculty of Sports Science, Yalova University, Yalova, Turkey

Background: Obesity, a health problem, is caused by insufficient physical activity and unhealthy nutrition. For individuals with obesity, non-weight-bearing and enjoyable exercises, such as Pilates, should be implemented as physical activity due to potential pain and injury. This study aims to examine the effects of an 8-week Pilates exercise program on body composition, posture, and psychosocial parameters in individuals with and without obesity. **Methods:** The study included 58 female participants: 30 individuals with obesity ($n_{\text{Pilates}} = 15$; $n_{\text{Control}} = 15$) and 28 individuals without obesity ($n_{\text{Pilates}} = 14$; $n_{\text{Control}} = 14$). Participants performed Pilates exercises twice a week for 8 weeks. Body mass index, body fat percentage, and posture scores were calculated, and from a psychosocial perspective Emotional Eating Scale, Social Physique Anxiety Inventory, and Self-Esteem Scales were administered before and after the 8 weeks. **Results:** While Pilates prevented an increase in body mass index and body fat percentage in women with obesity, it did not affect the women without obesity. Pilates improved posture scores in both groups. Additionally, Pilates reduced Emotional Eating and Social Physique Anxiety and increased Self-Esteem scores in women with obesity. No significant changes were observed in women without obesity. **Conclusion:** Pilates can be preferred to prevent increases in body mass index and body fat percentage in women with obesity, but detailed studies are needed regarding its effects on body composition. Pilates can be chosen to improve posture in women. Pilates can also strengthen choices to prevent obesity by increasing social adaptation through the psychosocial change it creates in women with obesity problems. (ClinicalTrials.gov Identifier: NCT06851026).

Keywords: exercise, physical activity, physical fitness, psychology, health promotion

Key Points

- Pilates training may improve body composition, postural alignment, and psychosocial well-being among women, supporting its role in health promotion programs.
- Distinct effects of Pilates were observed in women with and without obesity, highlighting the importance of tailored exercise interventions for diverse populations.
- This randomized controlled trial underscores Pilates as a feasible strategy to address physical and psychosocial health in women's preventive care.

Obesity is a chronic disease that occurs when the fat mass in the body exceeds the normal level due to the intake of more energy than is expended.¹ Obesity, which is among the 4 main metabolic variables that increase the risk of noncommunicable diseases, can affect the functioning of all organs and systems particularly the endocrine and cardiovascular systems, leading to health problems and death.^{1,2} The prevalence of obesity among adults worldwide has more than doubled between 1990 and 2022, reaching 16% in 2022.¹ In Turkey, according to data from the Turkey Health Research conducted by the Turkish Statistical Institute, obesity rates among women increased by 32.5% between 2008 and 2014.³ The main causes of this include insufficient physical activity and poor nutrition.⁴ Especially, the participation rates of women in physical activities are lower compared with men.⁵ Increasing

physical activity in women can help prevent obesity. In fact, the basis of obesity treatment is lifestyle changes aimed at weight management.⁶ Exercise therapy is the most commonly used approach, as regular exercise leads to changes in body composition and increases the effectiveness of diets.^{7,8} Studies show that endurance, resistance, and high-intensity interval training are effective exercise modalities in treating obesity.^{9,10} Due to the potential for pain and lower dependency on traditional exercise types among individuals with obesity,^{5,11} non-weight-bearing and enjoyable exercises should be preferred for these individuals.¹² Pilates, which includes various exercises, does not require high costs, prevents monotony, does not require weights, and can be customized, reducing the potential for pain and thus promoting physical development and regulation of body composition.^{11–13} Moreover, Pilates exercises focus on lumbopelvic stabilization; stability; segmental mobilization of the spine, shoulder, elbow, hip, knee, ankle mobilization and stability; coordination; and balance.¹⁴ For these reasons, it can be seen as a preferable option for individuals with obesity.^{15,16}

Delice  <https://orcid.org/0009-0008-2993-6378>

Arslan Kabasakal (sema.kabasakal@yalova.edu.tr) is corresponding author,
 <https://orcid.org/0000-0002-4552-9640>

The effects of Pilates on body mass index (BMI) and fat percentage are controversial. Saltan¹⁷ found that Pilates exercises had a positive effect on posture and BMI in women. In middle-aged sedentary women, Pilates was found to have an effect on reducing BMI, fat mass, and increasing muscle mass.¹⁸ Positive effects of Pilates on body composition, BMI, and some anthropometric variables were also observed in sedentary women.¹⁹ There are also studies showing that Pilates exercises have no effect on body composition.^{20,21} Most studies on women with obesity have focused on the effects of Pilates on body composition.^{15,22} However, there are no studies determining the psychological effects of Pilates on individuals with obesity.²² A study by Korkmaz et al²³ found that Pilates had positive effects on social appearance anxiety and self-esteem (SE) in individuals. There were no studies found in the literature examining the effects of Pilates exercises on emotional eating (EE) and posture. Additionally, there are no studies examining the effects of Pilates on posture in individuals with obesity.

Most studies in the literature have focused on sedentary women; however, no study has specifically examined the effects of Pilates-based exercises on individuals with and without obesity. This study aims to investigate the effects of an 8-week Pilates intervention on postural alignment, BMI, fat percentage, and psychosocial parameters (EE, social physical anxiety, and SE) in individuals with and without obesity. Thus, this study will provide valuable insights into the role of Pilates in weight management, posture correction, and psychosocial well-being among individuals with and without obesity.

Methods

Sample Size

The computer program G*Power 3.1 (Heinrich Heine University Düsseldorf, Düsseldorf, Germany) was used to determine the sample size. In this program, sample size calculation was made for the analysis of variance (ANOVA; within-between interaction) test, which examines the interaction of changes over time (pre–post differences) between 4 groups. Based on the analysis, a minimum required sample size of 56 participants was calculated, considering a medium expected Cohen *f* effect size of 0.25, a significance level of 0.05, and a desired power of 0.85. This effect size (Cohen *f* = 0.25) was determined based on Cohen widely accepted classification for medium effects in ANOVA analyses.²⁴ However, accounting for potential participant dropout, 60 participants who met the inclusion criteria were included in the study. Of these participants, 30 were women with obesity and 30 were women without obesity. Based on this analysis, 15 randomly selected individuals from each group were assigned to the Pilates exercise groups, while the remaining participants were placed in the control groups. Two participants from the group of women without obesity were unable to continue the study. As a result, the study was completed with 58 participants, consisting of 28 women without obesity and 30 women with obesity (Figure 1).

Participants

The study included 60 sedentary female participants with an average age of 35.64 (10.91) years, an average weight of 73.19 (16.38) kg, and an average height of 160.31 (5.67) cm. Among the participants, 30 were women with obesity, and the remaining 30 were women without obesity. Participants were randomly divided into 4 groups (Pilates with obesity, Pilates without obesity, control

with obesity, control without obesity) in accordance with the inclusion criteria. There were 15 participants in each group. Participants were randomly assigned to the intervention or control group using a lottery method, in which group names were placed in identical folded papers and drawn from a box by an independent staff member. Staff recruiting and assigning participants to interventions did not have access to the random allocation sequence. This study was designed as a double-blind randomized controlled trial. Participants were blinded to their obesity classification, meaning they were unaware of whether they were assigned to the obesity or nonobesity group. Additionally, the researcher who administered the Pilates interventions was blinded to participants' group assignments and obesity status. This ensured that both participants and interventionists were masked, reducing the risk of performance bias.

Two participants from the group of women without obesity could not continue the study and were excluded, resulting in a final sample of 28 women without obesity and 30 women with obesity (Figure 1). The inclusion and exclusion criteria for the study are as follows:

Inclusion Criteria

- Being a sedentary individual and not doing regular physical activity.
- Being 20 years old or older.
- Having a BMI of 30 kg/m² or above for inclusion in the group of women with obesity.
- Having a BMI below 30 kg/m² for inclusion in the group of women without obesity.
- Not having any health problems that would prevent participation in exercise.
- Not having undergone surgery affecting the musculoskeletal system within the last year.

Exclusion Criteria

- Voluntary withdrawal from the study.
- Failure to comply with the required procedures of the study.
- Arbitrary disruption of the exercise schedule more than 2 times.

Study Design

This clinical trial followed Consolidated Standards of Reporting Trials (CONSORT) 2025 guidelines. The study was designed as a randomized controlled trial. This study has been registered with the Clinical Trials Registry (NCT06851026).

The study was conducted at Yalova University Faculty of Sports Sciences. The study consists of 3 phases. In the first phase, individuals were invited to the measurement area and informed about the study. Subsequently, participants completed the Informed Consent Form and Participant Evaluation Form based on voluntary participation. The participant information form inquired about any health problems the participants might have. Then, participants underwent postural assessments, and measurements of weight, height, BMI, and skinfold thickness (ST) were taken. Additionally, participants completed the SE Scale, Social Physique Anxiety (SPA) Scale, and EE Scale. Participants were divided into 2 groups: the exercise group and the control group.

In the second phase, the exercise groups performed Pilates-based exercises twice a week for 8 weeks. The exercise duration was chosen as 8 weeks because studies examining exercise effects

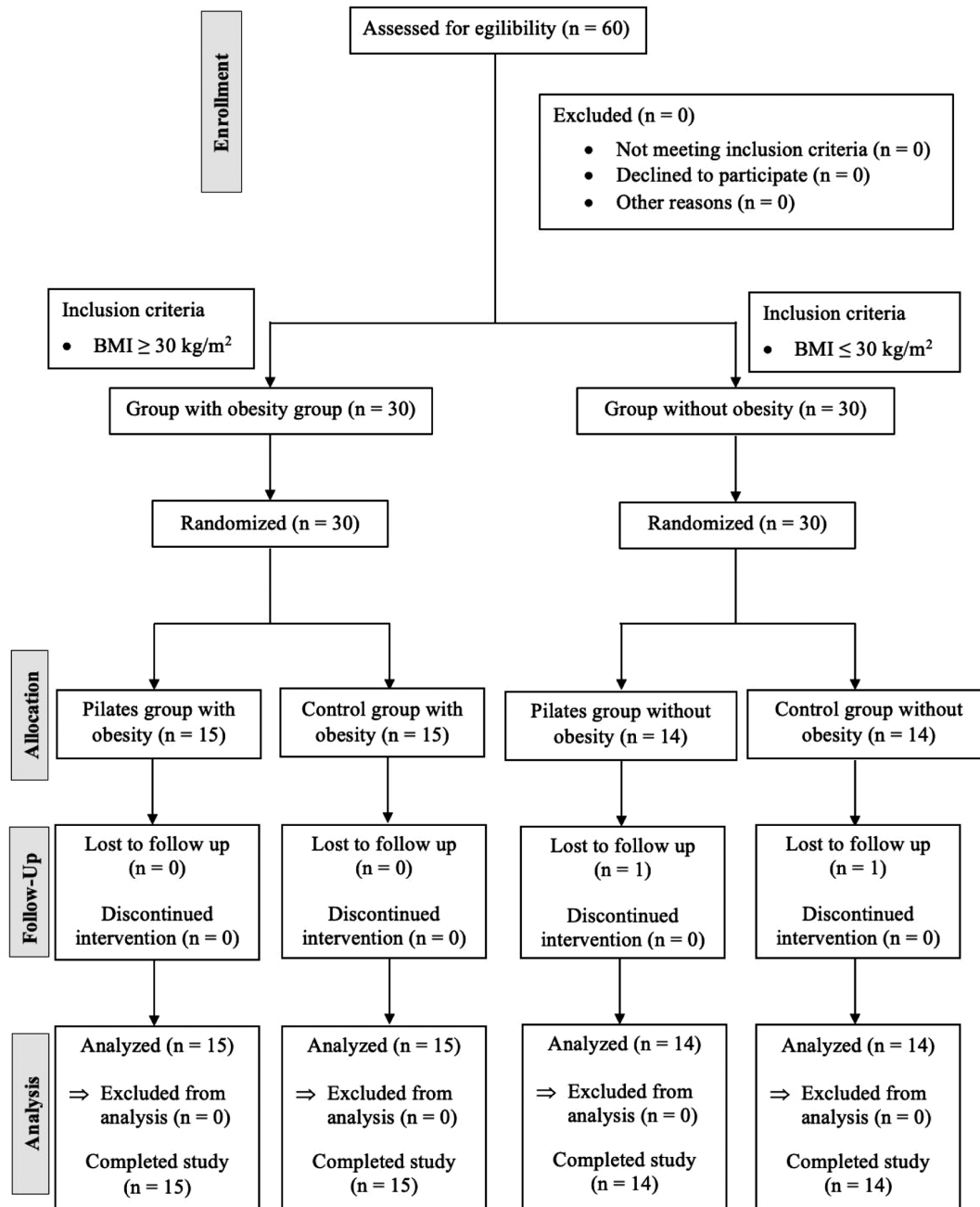


Figure 1 — CONSORT (Consolidated Standards of Reporting Trials) flow diagram.

in the literature mostly implemented and applied an 8-week exercise protocol.¹⁹

In the third phase, at the end of the eighth week, participants completed the scales again, and their measurements were taken. The results obtained were used to determine how the outcomes varied between women with and without obesity due to the exercise and whether there was any difference between the 2 groups.

Body Composition Measurements

BMI Calculation

Weight was measured using a calibrated digital scale, with participants wearing light clothing and no shoes to ensure accuracy.

Height was measured using a wall-mounted stadiometer, with participants standing upright, feet together, and heels against the wall. The calculated BMI values were used to create participant groups. Individuals with a BMI over 30 kg/m² were categorized as women with obesity, while those with a BMI below 30 kg/m² were categorized as women without obesity.²⁵

Skinfold Thickness Measurement and Body Fat Percentage Calculation

In studies related to ST measurements in the literature, measurements were taken at the triceps, marked from a point between the shoulder and elbow. The subscapular ST was marked at the inferior angle of the left scapula bone. The biceps ST was measured over

the cubital fossa at the level of the triceps skinfold measurement point. The suprailiac ST was measured at the midpoint of the distance between the iliac bone and the lowest rib.²⁶ Using the measurements taken, along with the age, body weight, and gender of the individuals, body density was calculated using the equation prepared by Durnin and Womersley.²⁷ After calculating body density, the value was inserted into the Siri equation to calculate body fat percentage (BFP).²⁶

New York Posture Analysis

This assessment is conducted by scoring potential postural changes in 13 different parts of the body. If the person's posture is correct, a score of 5 is given; if moderately impaired, a score of 3; and if severely impaired, a score of 1. The total score ranged from a maximum of 65 to a minimum of 13. A higher score indicates better posture alignment.²⁸ Analyses were conducted based on the total New York Posture Analysis Score.

Psychosocial Inventories

Emotional Eating Scale

This scale is used to determine individuals' EE behaviors. It was developed by Garaulet et al.²⁹ The Turkish translation and validity-reliability study of the scale was conducted by Arslantaş et al.³⁰ The scale consists of 10 items and is structured in a 4-point Likert format (0 = Never, 1 = Sometimes, 2 = Often, 3 = Always). The scale has no reverse items and is scored between 0 and 30. A score of "30" indicates the highest level of EE behavior, while a score of "0" indicates the absence of EE behavior. In the study, analyzes were made based on total scores. The total Cronbach alpha internal consistency coefficient of the scale was .86, indicating that the scale was a valid and reliable measurement tool.

Social Physique Anxiety Inventory

The Social Physique Anxiety Inventory is a 12-item scale developed by Hart et al.³¹ in 1989 to determine individuals' levels of SPA. The Turkish adaptation study was conducted by Ballı and Aşçı³² in 2006. The scale is a 5-point Likert structure (completely false, mostly false, sometimes false, sometimes true, mostly true, and completely true). The minimum possible score from the inventory is 12, and the maximum is 60. Higher scores indicate higher levels of anxiety about physical appearance. Items 1, 2, 5, 8, and 11 are reverse scored. In the study, analyzes were made based on total scores. The total Cronbach alpha internal consistency

coefficient of the scale was .85, indicating that the scale was a valid and reliable measurement tool.

Self-Esteem Scale

The Rosenberg SE Scale, developed by Arıcak³³ in 1999, was used in this study. This 5-point Likert scale allows participants to indicate their agreement with statements as "strongly agree," "agree," "neutral," "disagree," and "strongly disagree." In the study, analyzes were made based on total scores. The total Cronbach alpha internal consistency coefficient of the scale was .95, indicating that the scale was a valid and reliable measurement tool.

Pilates Exercises

Participants were given Pilates sessions twice a week for 8 weeks. Each session lasted approximately 40 minutes and consisted of a warm-up period, a main Pilates exercise period, and a cooldown period. The cooldown period consisted of a 5-minute low-level jog. Pilates exercises were selected to work the arm, trunk, back, and leg areas and were divided into 4 stages according to difficulty level (Table 1). Each stage was applied for 2 weeks. In these stages, the number of repetitions and sets was kept at 2 sets of 10 repetitions in the first week of each stage, while it was increased to 2 sets of 20 repetitions in the second week. The warm-up program and Pilates exercises to be applied are listed in Table 1.

Ethical Consideration

Ethics committee permission for the study was received from Yalova University Human Research Ethics Committee on December 07, 2023 with protocol number 2023/161. Participants were informed about the study and filled out an informed consent form. Participants were included in the study on a voluntary basis. The current research is in accordance with the Declaration of Helsinki.

Statistical Analysis

SPSS 26 package program was used to evaluate the data. The normality of the data was checked with the Shapiro–Wilk test, and skewness and kurtosis values. The normality was tested separately for BMI, fat percentage, posture score, EE, social physical anxiety, and SE variables. Since the skewness and kurtosis values were in the range of -2.0 to $+2.0$, the data show normal distribution.^{34,35} Mean and SD values were used for descriptive statistics. A 1-way ANOVA test was applied to compare the groups' demographic

Table 1 Warm-Up and Pilates Exercises Program

Warm-up protocol	Pilates exercises program			
	1–2 wk	3–4 wk	5–6 wk	7–8 wk
Breathing	T press	Biceps curls with band	Arm abduction with band	Plank
Imprint and release	Hip extension one leg	Tricep press with band	Standing flies with band	Bicycle kicks with the band
Hip rolls	Single-leg kicks	Flutter kicks	Russian twist	Bird dog
Spinal rotation	Swan prep	Toe taps	Scissor crunches	Swimming
Cat stretch	Child's pose	Glute bridge leg lifts	Kneeling arm and leg reach	Spine stretch
Scapula isolation	Pelvic curl	Chest lift	Cat-cow	Plank leg pulses
Arm circle	Side kicks	Side lying leg circles	Single-leg kicks	Double-leg stretch
			Plie squat	Front lunge

characteristics and pretest results, thus examining the success of the randomization. A mixed variance analysis “group (4)” $\times$ “time (2)” was used to analyze the effects of the 8-week intervention on each dependent parameter. The group factor consists of “Pilates group with obesity,” “Pilates group without obesity,” “control group with obesity,” and “control group without obesity” groups. The time factor consists of 2 factors: “pretest” and “posttest.” Dependent variables are BMI value, fat percentage, posture score, EE total score, social physical anxiety total scores, SE total scores. Bonferroni post hoc test was applied to determine within-group change. Additionally, analysis of covariance (ANCOVA) was employed to account for baseline differences in parameters that showed pretest differences according to the 1-way ANOVA results. In this analysis, pretest results were used as covariates, and the Bonferroni correction was applied for post hoc tests. An independent samples *t* test was also conducted to compare the changes observed after 8 weeks between women with and without obesity who participated in Pilates. Additionally, a chi-square analysis was conducted to examine changes in BMI categorization. Statistically, the significance value was accepted as $P < .05$. The effect of the mean score difference between the variables was evaluated with the Cohen *d* and eta squared (η^2) value. When Cohen *d* was in the range of “0.20 to 0.49,” it was considered a small effect, when it was “0.50 to 0.79,” it was considered a medium effect, and when it was “ >0.80 ,” it was considered a large effect. The η^2 value range was considered as “.01—small effect power, .06—medium effect power, .14 and above—large effect power.”³⁶

Results

Comparison of Demographic and Pretest Results Between Groups

In Table 2, the comparison of demographic variables and pretest results between groups was examined with the 1-way ANOVA test to see whether the random distribution of groups was successful. The 1-way ANOVA test showed no differences between groups for age, height, postural scores, EE scores, and SE scores (for age: $F_{3,54} = 0.41$, $P = .746$, $\eta^2 = .022$; for height: $F_{3,54} = 0.71$, $P = .549$, $\eta^2 = .038$; for postural scores: $F_{3,54} = 2.40$, $P = .078$, $\eta^2 = .118$; for EE scores: $F_{3,54} = 1.46$, $P = .235$, $\eta^2 = .075$; for SE scores: $F_{3,54} = 0.48$, $P = .694$, $\eta^2 = .026$). These findings suggest that randomization successfully balanced these variables across the groups. However, significant differences were found in weight and BFP between the 4 groups (for weight: $F_{3,54} = 56.53$, $P = .000$, $\eta^2 = .758$; for BMI: $F_{3,54} = 122.34$, $P = .000$, $\eta^2 = .872$; BFP: $F_{3,54} = 27.89$, $P = .000$, $\eta^2 = .608$). Post hoc analyses revealed that the Pilates and control groups with obesity had significantly higher weight, BMI, and BFP compared with the nonobesity groups ($P = .000$). When looking at the pretest SPA scores, significant differences were found between the groups ($F_{3,54} = 5.74$, $P = .002$, $\eta^2 = .242$). Post hoc test results indicate that Pilates and control group with obesity have significantly higher SPA scores compared to control group without obesity. Given the differences in weight, BFP, and BMI, this result is expected. However, the eta-squared

Table 2 Comparison of Demographic and Pretest Variables Between Groups

Variables		Sum of squares	df	Mean square	F	P	η^2	Post hoc test
Age, y	Between groups	151.16	3	50.39	0.41	.746	.022	$P < .05$
	Within groups	6632.24	54	122.82				
Weight, kg	Between groups	11,603.14	3	3867.71	56.53	.000*	.758	Group 1 > group 2, $P < .001$; Group 1 > group 4, $P < .001$; Group 3 > group 2, $P < .001$; Group 3 > group 4, $P < .001$
	Within groups	3694.59	54	68.42				
Height, cm	Between groups	69.71	3	23.24	0.71	.549	.038	$P > .05$
	Within groups	1762.28	54	32.63				
BMI, kg/m ²	Between groups	1941.54	3	647.18	122.34	.000*	.872	Group 1 > group 2, $P < .001$; Group 1 > group 4, $P < .001$; Group 3 > group 2, $P < .001$; Group 3 > group 4, $P < .001$
	Within groups	285.66	54	5.29				
BFP, %	Between groups	1829.54	3	609.85	27.89	.000*	.608	Group 1 > Group 2, $P < .001$; Group 1 > Group 4, $P < .001$; Group 3 > Group 2, $P < .001$; Group 3 > Group 4, $P < .001$
	Within groups	1180.84	54	21.87				
NPS-S	Between groups	530.58	3	176.86	2.40	.078	.118	$P > .05$
	Within groups	3974.52	54	73.60				
EE-TS	Between groups	110.97	3	36.99	1.46	.235	.075	$P > .05$
	Within groups	1366.62	54	25.31				
SE-TS	Between groups	486.13	3	162.04	0.48	.694	.026	$P > .05$
	Within groups	18,059.25	54	334.43				
SPA-TS	Between groups	823.96	3	274.65	5.74	.002**	.242	Group 1 > group 4, $P = .015^{**}$; Group 3 > group 4, $P = .014^{**}$
	Within groups	2585.69	54	47.88				

Abbreviations: BMI, body mass index; BFP, body fat percentile; EE-TS, emotional eating total scores; NPS-S, New York posture scale score; SE-TS, self-esteem total scores; SPA-TS, social physique anxiety total scores. Note: All data presented in the table are pretest results. Group 1, Pilates group with obesity; group 2, Pilates group without obesity; group 3, control group with obesity; group 4, control group without obesity.

* $P < .001$. ** $P < .05$.

value of 0.242 indicates that the impact of these variables on group homogeneity is relatively low.

Overall, these results highlight significant differences in weight, BMI, and BFP between groups with and without obesity. These outcomes are expected, as participants in the Pilates with obesity and control groups with obesity had a BMI over 30 kg/m², while those in the Pilates and control groups without obesity had a BMI below 30 kg/m². The lack of differences between the Pilates and control groups with obesity, as well as between the Pilates and control groups without obesity, suggests that the randomization process was successful.

ANCOVA analysis was applied to account for pretest differences in BMI, fat percentage, and SPA scores where there were differences between pretest results.

Body Mass Index Scores

When examining the BMI scores of the participants, significant differences were found between groups and in the interaction of time $\times$ group in Table 3 (for between groups: $F_{3,54} = 121.14$, $P = .000$, $\eta^2 = .871$; for time $\times$ group interaction: $F_{3,54} = 3.90$, $P = .014$, $\eta^2 = .178$). There was no significant difference in the pretest and posttest BMI scores of the participants regardless of the group variable ($F_{1,54} = 0.34$, $P > .05$, $\eta^2 = .006$). Post hoc tests revealed that the difference in the time $\times$ group interaction was between the pretest and posttest BMI values of the control group with obesity (mean difference [MD] = 0.73 [0.25], $P = .006$). It can be said that there was a significant increase in BMI over time in the control group with obesity. In the Pilates group with obesity, although not statistically significant, there was a decrease

in BMI values as seen in Table 4. This result indicates that while BMI values did not significantly change in women with obesity doing Pilates, BMI increased over time in those not doing Pilates.

When the effect of BMI pretest results was removed by the ANCOVA test, a significant difference was found between the posttest BMI values ($F_{1,53} = 278.69$, $P = .000$, $\eta^2 = .840$). Post hoc tests showed this difference between Pilates group with obesity and control group with obesity (MD = -1.19 [0.36], $P = .010$). Pilates exercises effectively decrease BMI in individuals with obesity regardless of BMI preliminary values. Looking at the data, while BMI values increased in the Pilates group without obesity, Pilates decreased in BMI values in the Pilates group with obesity (Table 3).

Furthermore, as shown in Table 5, changes in BMI differed between the Pilates groups with and without obesity ($t_{21,7} = -2.40$, $P = .030$, Cohen $d = 0.89$). While there was a decrease in BMI in the Pilates group with obesity, there was an increase in the Pilates group without obesity (Table 4).

In addition, BMI values were evaluated in 2 categories as obese (BMI < 30.00) and nonobese individuals (BMI $\geq$ 30.00) using chi-square analysis to determine whether there were any transitions between categories. According to the chi-square test results, there is a significant relationship between pre-Pilates and post-Pilates BMI categories ($\chi^2 [1, N = 58] = 54.13$, $P = .000$). This result shows that there was a significant change in BMI status after the intervention. In particular 96.7% of the participants who were obese at the beginning remained obese, only 3.3% moved out of the obesity category. All those who were not obese at the beginning maintained their BMI category (Table 6).

Table 3 Mixed Analysis of Variance Results

Variables	Group			Time			Group $\times$ time		
	<i>F</i>	<i>P</i>	η^2	<i>F</i>	<i>P</i>	η^2	<i>F</i>	<i>P</i>	η^2
BMI, kg/m ²	121.14	.000*	.871	0.34	.562	.006	3.90	.014*	.178
BFP, %	32.169	.000*	.641	14.594	.000*	.213	9.990	.000*	.213
NPS-S	2.43	.75	.119	44.67	.000*	.453	22.10	.000*	.551
EE-TS	2.70	.055	.130	21.340	.029*	.086	3.95	.013*	.180
SE-TS	0.459	.712	.025	1.145	.289	.021	3.55	.020*	.165
SPA-TS	6.187	.001*	.256	0.176	.677	.003	3.992	.012*	.182

Abbreviations: BFP, body fat percentile; BMI, body mass index; EE-TS, emotional eating total scores; NPS-S, New York posture scale score; SE-TS, self-esteem total scores; SPA-TS, social physique anxiety total scores.

* $P < .05$.

Table 4 Average Values of Participants' Pretest and Posttest Results

Variables	Pilates group with obesity		Control group with obesity		Pilates group without obesity		Control group without obesity	
	Pretest	Posttest	Pretest	Posttest	Pretest	Posttest	Pretest	Posttest
BMI, kg/m ²	34.17 (2.21)	33.70 (2.48)	34.02 (2.44)	34.75 (2.58)	22.94 (2.48)	23.08 (2.48)	22.12 (2.04)	22.03 (2.23)
BFP, %	42.65 (1.21)	41.69 (0.87)	44.48 (1.21)	44.59 (0.87)	32.40 (1.25)	34.76 (0.90)	32.42 (1.25)	39.64 (0.90)
NPS-S	39.53 (9.19)	49.93 (10.74)	39.27 (9.85)	37.80 (11.05)	38.07 (9.19)	53.79 (7.10)	45.93 (8.16)	45.21 (9.72)
EE-TS	12.60 (4.47)	9.53 (3.62)	14.87 (5.10)	15.00 (4.94)	11.07 (5.85)	10.57 (4.07)	12.21 (4.63)	12.21 (4.14)
SE-TS	117.13 (4.72)	124.33 (4.10)	114.13 (4.72)	114.67 (4.10)	118.43 (4.89)	120.57 (4.24)	122.21 (4.89)	117.79 (4.24)
SPA-TS	41.20 (1.79)	38.93 (1.37)	41.27 (1.79)	41.00 (1.37)	34.43 (1.85)	33.00 (1.42)	33.07 (1.85)	36.07 (1.42)

Abbreviations: BFP, body fat percentile; BMI, body mass index; EE-TS, emotional eating total scores; NPS-S, New York posture scale score; SE-TS, self-esteem total scores; SPA-TS, social physique anxiety total scores. Note: The values given in the table are in mean (SD).

Table 5 Comparison of Differences in Variables After Pilates in 2 Groups Doing Pilates

Variables	Groups	N	Mean	SD	t	df	P	Cohen d
BMI, kg/m ²	Pilates group with obesity	15	-0.47	0.85	-2.399	21.700	.026*	0.89
	Pilates group without obesity	14	0.14	0.46				
BFP, %	Pilates group with obesity	15	-0.96	1.55	-2.544	27	.017*	0.93
	Pilates group without obesity	14	2.37	4.83				
NPS-TS	Pilates group with obesity	15	10.40	7.84	-1.631	27	.114	0.60
	Pilates group without obesity	14	15.71	9.67				
EE-TS	Pilates group with obesity	15	-3.07	4.65	-1.771	27	.088	0.67
	Pilates group without obesity	14	-0.50	2.88				
SE-TS	Pilates group with obesity	15	7.20	8.68	1.595	27	.122	0.59
	Pilates group without obesity	14	2.14	8.37				
SPA-TS	Pilates group with obesity	15	-2.27	4.25	-.529	27	.601	0.20
	Pilates group without obesity	14	-1.43	4.27				

Abbreviations: BFP, body fat percentile; BMI, body mass index; EE-TS, emotional eating total scores; NPS-S, New York posture scale score; SE-TS, self-esteem total scores; SPA-TS, social physique anxiety total scores.

* $P < .05$.

Table 6 Chi-Square Test Results for Changes in Obesity Status From Pre- to Post-Pilates

BMI category	Post-Pilates			χ^2	P
	Individuals without obesity	Individuals with obesity	Total		
Pre-Pilates					
Individuals without obesity	28 (48.3%)	0 (0.0%)	28 (48.3%)	54.13	.000*
Individuals with obesity	1 (1.7%)	29 (50.0%)	30 (51.7%)		
Total	29 (50.0%)	29 (50.0%)	58 (100%)		

* $P < .05$.

Body Fat Percentage

When examining the fat percentages of the participants, significant differences were found in the time variable, between groups, and in the time $\times$ group interaction in Table 3 (for time variable: $F_{3,54} = 14.59$, $P = .000$, $\eta^2 = 0.213$; for between groups: $F_{1,54} = 32.17$, $P = .000$, $\eta^2 = .641$; for time $\times$ group interaction: $F_{3,54} = 9.99$, $P = .000$, $\eta^2 = .357$). Post hoc tests revealed significant differences in the pretest and posttest fat percentages in the Pilates group without obesity and the control group without obesity (for the Pilates group without obesity: MD = 2.37 (1.16), $P = .046$; for the control group without obesity: MD = 7.23 (1.16), $P = .000$).

ANCOVA test revealed a significant difference between posttest BFP values when the effect of pretest results was neglected ($F_{1,53} = 13.55$, $P = .001$, $\eta^2 = .204$). Post hoc tests showed that this difference was between the Pilates group without obesity and the control group with obesity (MD = -5.91 (1.55), $P = .002$), Pilates and control groups without obesity (MD = -4.87 (1.15), $P = .002$). In Table 2, it was observed that there was an increase in fat ratio in women without obesity depending on whether they did Pilates or not. Based on these results, it can be said that the increase in fat ratio is controlled by Pilates in women without obesity, and serious increases are prevented. In addition, there was a decrease in the fat ratio in the group with obesity who practiced Pilates, although it did not make a statistical difference (Table 2).

There was also a difference in the change in fat percentage between the Pilates groups with and without obesity ($t_{27} = -2.54$, $P = .017$, Cohen $d = 0.93$). This difference was in the direction of a greater reduction in fat percentage in the Pilates group with obesity (Table 5).

New York Posture Scale Scores

Significant differences were found in the time variable and the time $\times$ group interaction in the participants' posture scores in Table 3 (for time variable: $F_{1,54} = 44.67$, $P = .000$, $\eta^2 = .453$; for time $\times$ group interaction: $F_{3,54} = 22.10$, $P = .000$, $\eta^2 = .551$). However, no significant difference was found between groups independent of time in New York Posture Scale scores ($F_{3,54} = 2.43$, $P > .05$, $\eta^2 = .119$). Post hoc tests showed significant differences in the pretest and posttest in the Pilates group with obesity and the Pilates group without obesity (for the Pilates group with obesity: MD = 10.40 (1.76), $P = .000$; for the Pilates group without obesity: MD = 15.71 (1.82), $P = .000$). There was a significant increase in posture scores in both Pilates groups (Table 3). No significant difference was found in the change in posture scores between the 2 groups in Table 5 ($t_{27} = -1.63$, $P > .05$, Cohen $d = 0.60$). This result shows that the effect of Pilates exercises on posture was similar in both populations.

Emotional Eating Total Scores

Significant differences were found in the time variable and the time $\times$ group interaction in the participants' EE total scores in Table 5 (for time variable: $F_{1,54} = 5.05$, $P = .029$, $\eta^2 = .086$; for time $\times$ group interaction: $F_{3,54} = 3.95$, $P = .013$, $\eta^2 = .18$). No significant difference was found between groups independent of time in EE levels ($F_{3,54} = 2.70$, $P > .05$, $\eta^2 = .13$). Post hoc tests showed that the difference in the time $\times$ group interaction was due to the difference between the pretest and posttest of the Pilates group with obesity (MD = -3.07 (0.75), $P = .000$). There was a significant

decrease in EE levels after Pilates exercises in the Pilates group with obesity (Tables 3 and 4). No significant difference was found in the change in EE scores between the Pilates groups with and without obesity in Table 5 ($t_{27} = -1.77$, $P > .05$, Cohen $d = 0.67$).

Social Physique Anxiety Total Scores

Significant differences were found between groups and in the time $\times$ group interaction in the participants' SPA scores in Table 3 (for between groups: $F_{3,54} = 6.19$, $P = .001$, $\eta^2 = .256$; for time $\times$ group interaction: $F_{3,54} = 3.99$, $P = .012$, $\eta^2 = .182$). No significant difference was found in SPA scores in the time variable independent of the group ($F_{1,54} = 0.176$, $P > .05$, $\eta^2 = .003$). Post hoc tests showed that the difference in the time $\times$ group interaction was due to the difference between the pretest and posttest in the Pilates group with obesity and the control group without obesity (for the Pilates group with obesity: MD = -2.27 (1.13), $P = .049$; for the control group without obesity: MD = 3.00 (1.17), $P = .013$). There was a significant decrease in SPA scores over time in the Pilates group with obesity and a significant increase in the control group without obesity (Tables 3 and 4).

When the effect of the participants' pretest SPA scores was removed with the ANCOVA test, a significant change was found between the posttest SPA scores ($F_{1,53} = 80.17$, $P = .000$, $\eta^2 = .602$). Post hoc tests showed that this difference was between the Pilates group without obesity and control group with obesity (MD = -3.93 (1.34), $P = .029$) and between the Pilates group without obesity and control group without obesity (MD = -3.88 (1.28), $P = .023$). This result shows that, independent of the pretest SPA scores, the Pilates group without obesity had a decrease in social anxiety compared with the control with, and, without obesity. In other words, Pilates exercises are effective in reducing the anxiety felt about physical appearance in individuals without obesity.

No significant difference was found in the change in SPA scores between the Pilates groups with and without obesity in Table 5 ($t_{27} = -.53$, $P > .05$, Cohen $d = 0.20$). This result shows that Pilates exercises have similar effects in reducing SPA in individuals with and without obesity.

Self-Esteem Total Scores

Significant differences were found in the time $\times$ group interaction in the participants' SE total scores in Table 3 ($F_{3,54} = 3.55$, $P = .020$, $\eta^2 = .165$). However, no significant difference was found in SE scores in the time variable independent of the group and between groups independent of time (for time variable: $F_{1,54} = 1.15$, $P > .05$, $\eta^2 = .021$; for between groups: $F_{1,54} = 0.46$, $P > .05$, $\eta^2 = .025$). Post hoc tests showed that the difference in the time $\times$ group interaction was due to the difference between the pretest and posttest of the Pilates group with obesity (MD = 7.20 (2.50), $P = .006$). There was a significant increase in SE levels after Pilates exercises in the Pilates group with obesity (Tables 3 and 4). No significant difference was found in the change in SE scores between the Pilates groups with and without obesity in Table 5 ($t_{27} = 1.60$, $P > .05$, Cohen $d = 0.59$).

Discussion

Pilates is a type of exercise that promotes physical health and motor functions in women.^{37–39} The aim of the study was to determine the effects of 8 weeks of Pilates exercises on posture, fat percentage, BMI

values, EE, SE, and SPA levels in women with and without obesity and to determine whether it is more effective in women with obesity.

The effects of Pilates on body composition have been examined in various studies, revealing inconsistencies. Arguz and İribalci⁴⁰ found that Pilates exercises led to reductions in BFP and BMI values in women. Pereira et al⁴¹ suggested that Pilates is an effective exercise type for preventing obesity. However, there are studies indicating that Pilates improves fat and lean mass but does not affect BMI values.^{42,43} Conversely, some studies have found no effect of Pilates on body composition.^{10,11} This study suggests that Pilates may help prevent BMI increases and reduces BMI value in women with obesity who do not engage in physical activity, though longer intervention durations might yield more significant effects. One of the limitations of the study is that the exercise period was limited to 8 weeks. It is recommended that future studies examine the effects of long-term Pilates practices on obese individuals. Additionally, Pilates was more effective in reducing BMI in women with obesity than in those without, indicating its potential as a preventive measure against obesity progression. Moreover, an increase in BFP over time was observed in nonobese women regardless of Pilates participation; however, Pilates appeared to reduce the magnitude of this increase. In obese individuals, Pilates contributed to a reduction in BFP. To achieve greater levels of weight loss, it may be necessary to incorporate dietary interventions alongside Pilates.^{44,45} Therefore, future studies should investigate the effects of combining Pilates with dietary interventions on obesity.

Pilates exercises are known to reduce body asymmetries and improve postural alignment in women.^{46–49} This is achieved by enhancing muscle control and developing core stabilization.⁵⁰ Studies involving individuals with obesity have shown improvements in physical fitness levels.^{16,51} However, there is no direct study on the effect of Pilates on posture in individuals with obesity. This study found that Pilates improved posture in both women with and without obesity. Additionally, an 8-week Pilates program led to similar levels of improvement in posture in both groups. Given that the literature supports Pilates' role in improving postural alignment in women, the study results are consistent with existing literature. The current study reveals a novel finding that Pilates significantly enhances postural alignment in women with obesity, likely due to increased physical fitness and muscle strength balance provided by the Pilates exercises. However, more studies are needed in the literature examining the effect on posture in obese individuals because the current findings offer a new contribution on this subject.

Pilates exercises increase the level of psychological health as well as physical health in women.⁵² The current study also investigated the effects of Pilates on psychosocial variables. The psychosocial parameters included EE, SPA, and SE. EE is characterized by an increase in food intake in response to negative emotional states. Increased energy intake associated with EE complicates weight control and weight loss.⁵³ It is already known to be related to high BMI and associated with risk for obesity-related diseases.⁵⁴ Moreover, physical activity has been shown to improve eating behavior.⁵³ Although no prior studies have explored the effects of Pilates on EE in individuals with obesity, this study found that Pilates prevented increases in EE levels in women with obesity, potentially promoting controlled eating and reductions in body fat. However, it had no impact on EE levels in women without obesity.

SPA refers to the social anxiety experienced concerning one's physical appearance when one believes, or, perceives that others evaluate it negatively.³¹ Studies have shown that obesity is associated with higher levels of SPA compared with individuals of normal weight.^{55,56} Pilates exercises have been found to increase

psychological well-being by reducing SPA in women.^{57–59} This study observed significant reductions in SPA in women with obesity following Pilates, while similar effects were noted in women without obesity. Time-related increases in SPA were observed in the control groups, further highlighting Pilates' positive influence.

Finally, SE is associated with obesity, and low SE is one of the factors that increases the risk of obesity.⁶⁰ Additionally, SE helps in maintaining a healthy weight.⁶¹ Interventions that promote healthy behaviors and lifestyles are needed to enhance SE.⁶² Literature indicates that Pilates enhances SE in women.^{63–65} However, its effects on women with obesity have not been established. This study found that Pilates significantly increased SE in women with obesity, potentially due to improvements in posture and reductions in body fat. Similar effects were observed in women without obesity, where Pilates enhanced SE, while the control groups experienced declines.

In conclusion, this study supports the positive effects of Pilates on physical and psychological well-being in women with obesity. The findings highlight Pilates' role in preventing BMI increases, reducing EE level improving posture, and enhancing SE and SPA. Future research should focus on long-term effects and the potential benefits of combining Pilates with dietary interventions for sustained improvements in obesity management.

Limitations and Recommendations

One of the most notable limitations of this study is the small sample size particularly relevant given the 4-group design, were included per group. This limitation impacts the ability to detect significant effects and affects the representativeness of the findings. Additionally, the study's findings may be influenced by the lack of control and standardization of participants' diets during the 8-week Pilates intervention. Future research should control and standardize dietary intake to eliminate the impact of nutrition and isolate the effects of Pilates on body composition. The 8-week duration of the Pilates program may also limit the observed effects on body composition. Future studies should explore the effects of long-term Pilates interventions. Based on these findings, Pilates is recommended as an exercise modality for the prevention of obesity, while recognizing the need for further research to address the limitations identified in this study.

Conclusions

The current study highlights that, in the absence of any intervention, women with obesity experience a time-dependent increase in BMI and BFP. Although Pilates can mitigate this increase, statistically significant reductions were not observed. In addition, Pilates did not prevent the increase in BMI and BFP among women without obesity. Importantly, Pilates has been found to improve posture in both women with and without obesity, with significant postural improvements observed in women with obesity. Furthermore, Pilates led to reductions in EE and SPA while increasing SE among women with obesity. The reduction in EE may facilitate better control over dietary habits, contributing to improvements in body composition. Enhanced SE and decreased SPA may further promote social engagement and higher physical activity levels in this population.

Acknowledgments

Funding Source/Trial Registration: This study was supported by TÜBİTAK Scientist Support Department under project 2209-A, number

1919B012224567. We would like to thank TÜBİTAK for its support in the creation of this study. The trial was registered at ClinicalTrials.gov (Identifier: NCT06851026).

References

1. World Health Organization. (2024). *Obesity and overweight*. Accessed June 24, 2024, from <https://www.who.int/en/news-room/fact-sheets/detail/obesity-and-overweight>
2. World Health Organization. (2023). *Noncommunicable diseases*. Accessed June 24, 2024, from <https://www.who.int/zh/news-room/fact-sheets/detail/noncommunicable-diseases>
3. Çukur A, Arı Erdem İ. The role of obesity taxes in the fight against obesity: an evaluation for Turkey. *J Turk Court Accounts*. 2017; 10(106):121–146.
4. Ross SE, Flynn JJ, Pate RR. What is really causing the obesity epidemic? A review of reviews in children and adults. *J Sports Sci*. 2016;34(12):1148–1153. doi:10.1080/02640414.2015.1093650
5. Reece LJ, Foley BC, McInerney C, Bellew B, Bauman AE. *Women and Girls in Sport, Active Recreation and Physical Activity—A Participation Review*. Springer Group; 2017, The University of Sydney. ISBN: 978-1-74210-440-9. <https://www.sport.nsw.gov.au/sites/default/files/2021-04/women-in-sport-sprinter-evidence-review.pdf>
6. Pierzak-Stepień M. Available therapies for the treatment of obesity. Are lifestyle modifications and pharmacotherapy enough to reduce weight?—Review of the literature. *J Educ Health Sport*. 2023;44(1): 128–155. doi:10.12775/JEHS.2023.44.01.009
7. Wirth A, Wabitsch M, Hauner H. The prevention and treatment of obesity. *Dtsch Arztebl Int*. 2014;111(42):705. doi:10.3238/arztebl.2014.0705
8. Lambert DC, Kane J, Newberry C. Lifestyle therapy for obesity. *Gastrointest Endosc Clin*. 2024;34(4):577–589. doi:10.1016/j.giec.2024.03.003
9. Petridou A, Siopi A, Mougios V. Exercise in the management of obesity. *Metabolism*. 2019;92:163–169. doi:10.1016/j.metabol.2018.10.009
10. Li D, Chen P. The effects of different exercise modalities in the treatment of cardiometabolic risk factors in obese adolescents with sedentary behavior—a systematic review and meta-analysis of randomized controlled trials. *Children*. 2021;8(11):1062. doi:10.3390/children8111062
11. Burgess E, Hassmén P, Welvaert M, Pumpa KL. Behavioural treatment strategies improve adherence to lifestyle intervention programmes in adults with obesity: a systematic review and meta-analysis. *Clin Obes*. 2017;7(2):105–114. doi:10.1111/cob.12180
12. Rayes ABR, de Lira CAB, Viana RB, et al. The effects of Pilates vs. aerobic training on cardiorespiratory fitness, isokinetic muscular strength, body composition, and functional tasks outcomes for individuals who are overweight/obese: a clinical trial. *PeerJ*. 2019;7: e6022. doi:10.7717/peerj.6022
13. Souza Cavina APD, Junior EP, Machado AF, et al. Effects of the mat pilates method on body composition: systematic review with meta-analysis. *J Phys Act Health*. 2020;17(6):673–681. doi:10.1123/jpah.2019-0171
14. Smith K, Smith E. Integrating pilates-based core strengthening into older adult fitness programs: implications for practice. *Top Geriatr Rehabil*. 2005;21(1):57–67. doi:10.1097/00013614-200501000-00007
15. Wang Y, Chen Z, Wu Z, Ye X, Xu X. Pilates for overweight or obesity: a meta-analysis. *Front Physiol*. 2021;12:643455. doi:10.3389/fphys.2021.643455
16. Nasiri E, Ganji Harsini A, Arabi F, Samadi A, Rashki Kamak Z. Eight weeks of Pilates exercise improved physical performance of

- overweight and obese women without significant changes in body composition and serum myokines. *Sport Sci Health*. 2021;18(3):725–733. doi:[10.1007/s11332-021-00846-4](https://doi.org/10.1007/s11332-021-00846-4)
17. Saltan A. Investigation of the effects of pilates based exercise on posture, quality of life, depression symptoms and pain in women. *Firat Uni J Health Sci*. 2018;32(1):31–36.
 18. Bastık C, Cicioglu HI. Comparison of the effect of the mat and reformer pilates exercises on the waist-hips ratio and body compositions of the middle-aged sedentary women. *Prog Nutr*. 2020;22(suppl 1):50–57. doi:[10.23751/pn.v22i1-s.9784](https://doi.org/10.23751/pn.v22i1-s.9784)
 19. Uzun A, Demir B. Effect of pilates and reformer exercises on body composition. *Int J Appl Exerc Phys*. 2020;9(10):148–156.
 20. Sekendiz B, Altun Ö, Korkusuz F, Akin S. Effects of Pilates exercise on trunk strength, endurance and flexibility in sedentary adult females. *J Bodyw Mov Ther*. 2007;11(4):318–326. doi:[10.1016/j.jbmt.2006.12.002](https://doi.org/10.1016/j.jbmt.2006.12.002)
 21. Bergamin M, Gobbo S, Bullo V, et al. Effects of a Pilates exercise program on muscle strength, postural control and body composition: results from a pilot study in a group of post-menopausal women. *Age*. 2015;37(6):118. doi:[10.1007/s11357-015-9852-3](https://doi.org/10.1007/s11357-015-9852-3)
 22. Batrakoulis A. Psychophysiological adaptations to Pilates training in overweight and obese individuals: a topical review. *Disease*. 2022;10(4):71. doi:[10.3390/diseases10040071](https://doi.org/10.3390/diseases10040071)
 23. Korkmaz C, Çalık BB, Kabul EG. Investigation of the effects of clinical Pilates exercises on physical fitness and psychosocial parameters in healthy women. *J Exerc Ther Rehabil*. 2022;9(3):153–163. doi:[10.15437/jetr.1026091](https://doi.org/10.15437/jetr.1026091)
 24. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates; 1988.
 25. Bailey KV, Ferro-Luzzi A. Use of body mass index of adults in assessing individual and community nutritional status. *Bull World Health Organ*. 1995;73(5):673.
 26. Baysal A. Control of body weight. In: Baysal A, Bozkurt N, Pekcan G, Besler T, Aksoy M, Merdol TK, eds. *Diet Handbook (Turkish: Diyet El Kitabı)*. 7th ed. Hatipoğlu Publisher; 2002:39–63.
 27. McRoberts LB, Cloud RM, Black CM. Evaluation of the New York posture rating chart for assessing changes in postural alignment in a garment study. *Cloth Text Res J*. 2013;31(2):81–96. doi:[10.1177/0887302X13480558](https://doi.org/10.1177/0887302X13480558)
 28. Garaulet M, Canteras M, Morales E, et al. Validation of a questionnaire on emotional eating for use in cases of obesity; the Emotional Eater Questionnaire (EEQ). *Nutr Hosp*. 2012;27(2):645–651. doi:[10.1590/S0212-16112012000200043](https://doi.org/10.1590/S0212-16112012000200043)
 29. Arslantaş H, Dereboy F, Yüksel R, İnalkaç S. Validity and reliability of the Turkish version of the emotional eater questionnaire. *Turk J Psychiatry*. 2019;30:520. doi:[10.5080/u23520](https://doi.org/10.5080/u23520)
 30. Hart EA, Leary MR, Rejeski WJ. Tie measurement of social physique anxiety. *J Sport Exerc Psychol*. 1989;11(1):94–104. doi:[10.1123/jsep.11.1.94](https://doi.org/10.1123/jsep.11.1.94)
 31. Ballı ÖM, Aşçı FH. Reliability and validity of “social physique anxiety scale.” *Hacettepe J Sport Sci*. 2006;17(1):11–19.
 32. Arıca OT. *Self-Esteem and Vocational Self-Esteem Enhancement Through Group Counseling* [Doctoral thesis] Marmara University; 1999.
 33. George D, Mallery M. *SPSS for Windows Step by Step: A Simple Guide and Reference, 17.0 Update*. Pearson; 2010.
 34. Gürbüz S, Şahin F. *Research Methods in Social Sciences (Turkish: Sosyal Bilimlerde Araştırma Yöntemleri)*. Seçkin Publisher; 2018.
 35. Büyüköztürk Ş. *Manual of Data Analysis for Social Sciences (Turkish: Sosyal Bilimler için Veri Analizi El Kitabı)*. Pegem Academy Publisher; 2011.
 36. Lange C, Unnithan VB, Larkam E, Latta PM. Maximizing the benefits of Pilates-inspired exercise for learning functional motor skills. *J Bodyw Mov Ther*. 2000;4(2):99–108. doi:[10.1054/jbmt.1999.0161](https://doi.org/10.1054/jbmt.1999.0161)
 37. Latey P. The Pilates method: history and philosophy. *J Bodyw Mov Ther*. 2001;5(4):275–282. doi:[10.1054/jbmt.2001.0237](https://doi.org/10.1054/jbmt.2001.0237)
 38. Eyigor S, Karapolat H, Yesil H, et al. Effects of pilates exercises on functional capacity, flexibility, fatigue, depression and quality of life in female breast cancer patients: a randomized controlled study. *Eur J Phys Rehabil Med*. 2010;46(4):481–487.
 39. Arguz H, İribalçı Ş. The effect of pilates exercises applied to overweight and obese women on body composition. *Turk J Sports Med*. 2023;25(3):507–516. doi:[10.15314/tsed.1350628](https://doi.org/10.15314/tsed.1350628)
 40. Pereira MJ, Dias G, Mendes R, et al. Efficacy of Pilates in functional body composition: a systematic review. *Appl Sci*. 2022;12(15):7523. doi:[10.3390/app12157523](https://doi.org/10.3390/app12157523)
 41. Omidali Z, Taheri H, Asfarjani F, Bambaiech E, Marandi SM. The effect of pilates training on selective physiological and physical fitness in untrained females with overweight. *J Res Rehabil Sci*. 2012;8(1):180–191. doi:[10.22122/JRRS.V8I1.485](https://doi.org/10.22122/JRRS.V8I1.485)
 42. Fourie M, Gildenhuis GM, Shaw I, et al. Effects of a mat Pilates programme on body composition in elderly women. *West Indian Med J*. 2013;62(6):524–528. doi:[10.7727/wimj.2012.107](https://doi.org/10.7727/wimj.2012.107)
 43. Mayo MJ, Grantham JR, Balasekaran G. Exercise-induced weight loss preferentially reduces abdominal fat. *Med Sci Sports Exerc*. 2003;35(2):207–213. doi:[10.1249/01.MSS.0000048636.46744.01](https://doi.org/10.1249/01.MSS.0000048636.46744.01)
 44. Kaner G, Ayer Ç. The effect of a weight-loss diet in women doing reformer Pilates: a 12-week evaluation. *J Contemp Med*. 2022;12(1):19–26. doi:[10.16899/jcm.1004825](https://doi.org/10.16899/jcm.1004825)
 45. Cruz-Ferreira A, Fernandes J, Kuo YL, et al. Does pilates-based exercise improve postural alignment in adult women? *Women Health*. 2013;53(6):597–611. doi:[10.1080/03630242.2013.817505](https://doi.org/10.1080/03630242.2013.817505)
 46. Lee HT, Oh HO, Han HS, et al. Effect of mat Pilates exercise on postural alignment and body composition of middle-aged women. *J Physic Ther Sci*. 2016;28(6):1691–1695. doi:[10.1589/jpts.28.1691](https://doi.org/10.1589/jpts.28.1691)
 47. Navega MT, Furlanetto MG, Lorenzo DM, Morcelli MH, Tozim BM. Effect of the Mat Pilates method on postural balance and thoracic hyperkyphosis among elderly women: a randomized controlled trial. *Rev Bras Geriatr Gerontol*. 2016;19(3):465–472. doi:[10.1590/1809-98232016019.150022](https://doi.org/10.1590/1809-98232016019.150022)
 48. Kim D, Lim J. The effect of pilates core exercise on body alignment. *Arch J Orthop Sports Phys Ther*. 2020;16(1):15–24. doi:[10.24332/aospt.2020.16.1.03](https://doi.org/10.24332/aospt.2020.16.1.03)
 49. Parveen A, Kalra S, Jain S. Effects of Pilates on health and well-being of women: a systematic review. *Bull Fac Phys Ther*. 2023;28(1):128. doi:[10.1186/s43161-023-00128-9](https://doi.org/10.1186/s43161-023-00128-9)
 50. Shin HJ, Choi KE, Jung WS, Lee MG. Effects of a 12-week Pilates on body composition, daily living fitness, and indices of lifestyle disease in obese elderly women. *Korean J Sport Sci*. 2023;32(2):667–679. doi:[10.35159/kjss.2023.04.32.2.667](https://doi.org/10.35159/kjss.2023.04.32.2.667)
 51. Arslan Kabasakal S. The effects of a 6-week pilates exercises on quality of life, depression, and musculoskeletal disorders in menopausal women. *Eur Res J*. 2025;11(2):296–303. doi:[10.18621/eurj.1603630](https://doi.org/10.18621/eurj.1603630)
 52. Santos MVLD, Melo CMD, Rosa JPP, et al. Emotional eating is related to carbohydrate intake in active women. *Motriz: Revista de Educação Física*. 2016;22:346–352. doi:[10.1590/S1980-6574201600040020](https://doi.org/10.1590/S1980-6574201600040020)
 53. Ayyıldız F, Akbulut G, Ermumcu MŞK, Tek NA. (2023). Emotional and intuitive eating: an emerging approach to eating behaviours related to obesity. *J Nutr Sci*. 2023;12:e19. doi:[10.1017/jns.2023.11](https://doi.org/10.1017/jns.2023.11)

54. Ravesh SMHH. Analysis of the role of moderating socio-cultural pressure for weight loss and physical changes in obese and normal weight adolescents. *Phys Educ Sport Health Cult in Mod Soc*. 2020; 1(49):107–115. doi:[10.29038/2220-7481-2020-01-107-115](https://doi.org/10.29038/2220-7481-2020-01-107-115)
55. Jeanningros A, Baillot A, Corno G, et al. Validation of a virtual environment to induce state social physique anxiety in women with obesity and social physique anxiety. *J Clin Med*. 2023;12(18):6065. doi:[10.3390/jcm12186065](https://doi.org/10.3390/jcm12186065)
56. Doğru Y. Comparison of social physique anxiety of fitness and pilates exercise in women. *Turk J Sport Exerc*. 2019;21(3):411–414. doi:[10.15314/tsed.635500](https://doi.org/10.15314/tsed.635500)
57. Farzaneh A, Koushkie Jahromi M, Salesi M, Rostami R. The ameliorating impact of pilates training on anxiety and body mass index in menopausal symptoms. *Women's Health Bull*. 2020;7(4): 46–53. doi:[10.30476/whb.2020.88198.1084](https://doi.org/10.30476/whb.2020.88198.1084)
58. Duyan M, Ilkim M, Çelik T. The effect of social appearance anxiety on psychological well-being: a study on women doing regular pilates activities. *Pakistan J Med Health Sci*. 2022;16(2):797–801. doi:[10.53350/pjmhs22162797](https://doi.org/10.53350/pjmhs22162797)
59. Byth S, Frijters P, Beaton T. The relationship between obesity and self-esteem: longitudinal evidence from Australian adults. *Oxf Open Econ*. 2022;1:1–14. doi:[10.1093/ooec/odac009](https://doi.org/10.1093/ooec/odac009)
60. Liu E, Chang SH. Self-esteem and weight status of young adults: findings from a pilot study. *J Edu Health Promot*. 2022;11(1):263. doi:[10.4103/jehp.jehp_1195_21](https://doi.org/10.4103/jehp.jehp_1195_21)
61. Jumaily ZSAFA, Ibrahim RH. Correlation between self-esteem and obesity among high secondary schools in Mosul City. *Int J Health Sci*. 2022;6(S8):1561–1568. doi:[10.53730/ijhs.v6nS8.10790](https://doi.org/10.53730/ijhs.v6nS8.10790)
62. Özdenk S, Imamoglu M. The effects of pilates, step and zumba exercises on self-esteem, happiness and communication skill levels. *Asian J Educ Train*. 2019;5(2):369–373. doi:[10.20448/journal.522.2019.52.369.373](https://doi.org/10.20448/journal.522.2019.52.369.373)
63. Balaji E, Karthika Banu RA. A six week pilates mat exercises protocol on self efficacy and self esteem of school level basketball girls. *Bharathiar National J Phys Educ Sport Exerc Sci*. 2020;11(1): 25–31.
64. Fretta TDB, Boing L, Stein F, Santos LD, Guimarães ACDA. Improved self-esteem after mat Pilates method intervention in breast cancer women undergoing hormone therapy: randomized clinical trial pilot study. *Rev Bras Cineantropom Desempenho Hum*. 2021;23: e76311. doi:[10.1590/1980-0037.2021v23e76311](https://doi.org/10.1590/1980-0037.2021v23e76311)
65. Durnin JVGA, Womersley J. Body fat assessed from total body density and its estimation from skinfold thickness: Measurements on 481 men women aged from 16 to 72 years. *British J Nutri*. 1974; 32(1):77–97. doi:[10.1079/BJN19740060](https://doi.org/10.1079/BJN19740060)